Remark — In Theorem 3, we assume the platform can select any network $\mathbf{P}$ it desires through its recommendation algorithm. This is without loss of generality. If we assume the social network originally begins as an arbitrary island network in Section 4, and the platform can hide and amplify content across different links, the same result readily follows.

5.2 Comparative Statics on r_P: Divisiveness and Polarization

In Theorem 3, the threshold r_P fully summarizes the extent to which misinformation will spread virally on social media. We next perform comparative statics for this threshold to understand the conditions under which the platform will create a filter bubble and propagate misinformation.

Proposition 4. *The optimal reliability threshold r_P increases as message divisiveness and/or belief polarization increases.*

Proposition 4 mimics the conclusions of Proposition 3(a). As divisiveness or polarization increases, content is consumed more aggressively within echo chambers and scrutinized more aggressively outside of them. In these settings, filter bubbles become more advantageous to the platform, especially when the relevant content has low reliability. This is because communities with more extreme beliefs now feel more strongly about news in general, rarely second-guess politically-congruent news, but often doubt and dislike counter-attitudinal news. As a result, low-reliability content would have been quickly disliked and stopped without echo chambers, which would have discouraged its sharing even by agents inclined to believe it. In contrast, inside of the platform's filter bubbles, such content continues to spread virally.

Proposition 4 also provides a possible (albeit of course speculative) interpretation for why accelerating political polarization and identity politics in the last two decades may have come with more aggressive filter bubble algorithms from social media sites (Apprich et al. (2018)). As the recent documentary *The Social Dilemma* puts it: "The way to think about it is as 2.5 billion Truman Shows. Each person has their own reality with their own facts. Over time you have the false sense that everyone agrees with you because everyone in your news feed sounds just like you." Tellingly in this context, while Facebook cracked down on misinformation prior to the 2020 election in part due to political pressure, its algorithms have resumed promotion of misinformation in November and December of 2020: "...the measures [Facebook] could take to limit harmful content on the platform might also limit its growth: In experiments Facebook conducted last month, posts users regarded in surveys as 'bad for the world' tended to have a greater reach — and algorithmic changes that reduced the visibility of those posts also reduced users' engagement with the platform...".[19]

6 Regulation

Our analysis so far raises the natural question of what types of regulations might counter the viral spread of misinformation and platform choices leading to excessive ideological homophily. We

[19]See *Vanity Fair*:
https://www.vanityfair.com/news/2020/12/with-the-election-over-facebook-gets-back-to-spreading-misinformation
and also https://www.technologyreview.com/2021/03/11/1020600/facebook-responsible-ai-misinformation/.